

2Classical Limits

Buce-Ithon

12th August, 2025

Contents

1	$\lim_{n \rightarrow \infty} \frac{a^n}{n^k}$	3
2	$\lim_{n \rightarrow \infty} n^{\frac{1}{n}}$	3
3	Some repurposed variants	4
3.1	$\lim_{n \rightarrow +\infty} \sqrt[n]{n!} \dots\dots\dots$	4

$$1 \quad \lim_{n \rightarrow \infty} \frac{a^n}{n^k}$$

The problem is:

$$\lim_{n \rightarrow \infty} \frac{a^n}{n^k} \quad \text{where } a > 1, k > 0$$

Solution (Solution1). Based on **Binomial Theorem**
 $a^n = [a \triangleq 1 + \lambda] = 1 + n\lambda + \frac{n(n-1)}{2}\lambda^2 + \dots > \frac{n(n-1)}{2}\lambda^2$
also $n - 1 > \frac{n}{2}$, as $n > 2$, then $a^n > \frac{n^2}{4}(a - 1)^2$.
As $k = 1$, $\lim_{n \rightarrow \infty} \frac{a^n}{n} = \infty$, and as $k > 1$, $\lim_{n \rightarrow \infty} \frac{a^n}{n^k} = \lim_{n \rightarrow \infty} \left(\frac{(a^{\frac{1}{k}})^n}{n}\right)^k = \infty$.

Solution (Solution2). Based on **L'Hospital's Rule**
 $\lim_{n \rightarrow \infty} \frac{a^n}{n^k} = \lim_{n \rightarrow \infty} \frac{\ln a \cdot a^n}{k \cdot n^{k-1}} = \lim_{n \rightarrow \infty} \frac{(\ln a)^2 \cdot a^n}{k(k-1) \cdot n^{k-2}} = \dots = \lim_{n \rightarrow \infty} \frac{(\ln a)^k \cdot a^n}{k!} = \infty$.

Solution (Solution3). (Haven't found a totally different solution yet.)

$$2 \quad \lim_{n \rightarrow \infty} n^{\frac{1}{n}}$$

The problem is:

$$\lim_{n \rightarrow \infty} n^{\frac{1}{n}}$$

Solution (Solution1). Due to 1. $a^n > \frac{n^2}{4}(a - 1)^2$, let $a = n^{\frac{1}{n}}$, then $0 < n^{\frac{1}{n}} < \frac{2}{\sqrt{n}}$. Thus $\lim_{n \rightarrow \infty} n^{\frac{1}{n}} = 1$.

Solution (Solution2). Based on Mean Value Inequalities Chain(GM-AM): $n^{\frac{1}{n}} = (\sqrt{n} \cdot \sqrt{n} \cdot 1 \cdots 1)^{\frac{1}{n}} \leq \frac{2\sqrt{n} + n - 2}{n}$, then $\lim_{n \rightarrow \infty} n^{\frac{1}{n}} = 1$.

3 Some repurposed variants

3.1 $\lim_{n \rightarrow +\infty} \sqrt[n]{n!}$

The problem is:

$$\lim_{n \rightarrow +\infty} \sqrt[n]{n!}$$

Solution (Solution1). Based on **Stirling's Approximation**: $n! \sim \sqrt{2\pi n} \left(\frac{n}{e}\right)^n$, then $\sqrt[n]{n!} \sim \sqrt[n]{\sqrt{2\pi n} \left(\frac{n}{e}\right)^n} = \frac{n}{e} \cdot (2\pi n)^{\frac{1}{2n}}$. Thus $\lim_{n \rightarrow +\infty} \sqrt[n]{n!} = \infty$.

Solution (Solution2). $n! \geq \left(\frac{n}{2}\right)^{\frac{n}{2}}$ because half of the factors are at least $\frac{n}{2}$

Solution (Solution3). Consider Taylor series, $e^x \geq \frac{x^n}{n!}$, $\forall x \geq 0, n \in \mathbb{N}$, then $n! \geq \frac{x^n}{e^x}$, take $x = n$ problem solved.

Solution (Solution4). Based on **Stolze Cesaro Theorem**:
 $\lim_{n \rightarrow +\infty} \sqrt[n]{n!} = \lim_{n \rightarrow +\infty} \frac{n!^{\frac{1}{n}}}{1} = \lim_{n \rightarrow +\infty} \frac{\ln(n!)}{n} = \lim_{n \rightarrow +\infty} \frac{\ln(n) + \ln(n-1) + \dots + \ln(1)}{n}$,
then apply Stolze Cessaro Theorem, $\lim_{n \rightarrow +\infty} \frac{\ln(n) + \ln(n-1) + \dots + \ln(1)}{n} =$
 $\lim_{n \rightarrow +\infty} \frac{\ln(n+1)}{1} =$, then $\lim_{n \rightarrow +\infty} \sqrt[n]{n!} = \infty$.

Solution (Solution5). Using a multiplicative variant of Gauss' trick we get: $(n!)^2 = (1 \cdot n)(2 \cdot (n-1)) \dots ((n-1) \cdot 2)(n \cdot 1) \geq n^n$.